



Sutra

A local-first workspace for the way students actually work.

Notes · planning · homework · AP prep · review · college · life — woven into one private workspace.



PRIVATE · LOCAL-FIRST · STUDENT-BUILT

A student's day is scattered across a dozen tools

Notes app

where ideas live

School portal

where grades live

Calendar

where time lives

Task manager

where to-dos live

AI chatbot

where answers live

Study tools

where review lives

Docs & files

where work lives

20 browser tabs

where focus dies

The cost of fragmentation

- Constant context-switching — re-finding where each thing lives
- Information lost between apps that never talk to each other
- Studying disconnected from the deadlines and notes that drive it
- Missed due dates, weak organization, and little ownership of your own data

Point tools solve tasks — not continuity

Each category does one job well, then hands you off. The thread between planning, writing, studying, tracking, and recovering your work breaks at every boundary.

Generic notes app

Great at writing. Blind to your deadlines, classes, and study schedule.

Task manager

Tracks to-dos. Doesn't hold your notes, exams, or review.

AI chatbot

Answers questions. No memory of your workspace; your data leaves your control.

School LMS

Shows assignments. Read-only, per-school, not your personal system.

Study app

Drills flashcards. Disconnected from the material and the calendar.

Students don't need another silo. They need one thread through all of it.

One workspace. Every thread.

Sutra is Sanskrit for a thread — the line that holds separate beads into one piece.

A student operating system, concretely:

- One private workspace for your whole academic life
- Notes, deadlines, classes, studying, and planning that reference each other
- Context preserved instead of scattered across unrelated apps
- Optional AI assistance that reads your workspace — and only with your approval
- Your data on your device, portable in a single file, owned by you



We use the thread metaphor sparingly — inside the app, labels stay literal: Today, Notes, Homework, Timeline.

An honest snapshot of the product today

● Available Today ● In Progress ● Experimental ● Roadmap

● Available Today

- Notes + rich editor
- Today / Deadline Radar
- Timeline calendar
- Homework
- AP Study + Battle Plan
- Review (5 modes)
- College / Life / Work
- Local-first save + .sutra backup
- Themes + CSS overrides
- Privacy & CSP hardening

● In Progress

- Mobile QA across all surfaces
- Testing Hub depth
- Onboarding polish
- Plugin ecosystem
- Header-level security on host

● Experimental

- Cram surface
- Proactive assistance

● Roadmap

- Optional opt-in sync
- Service-worker offline
- Cross-device workflow
- Broader student-OS

How the workspace fits together

Repository Module Map

Where the code lives.



Repository module map

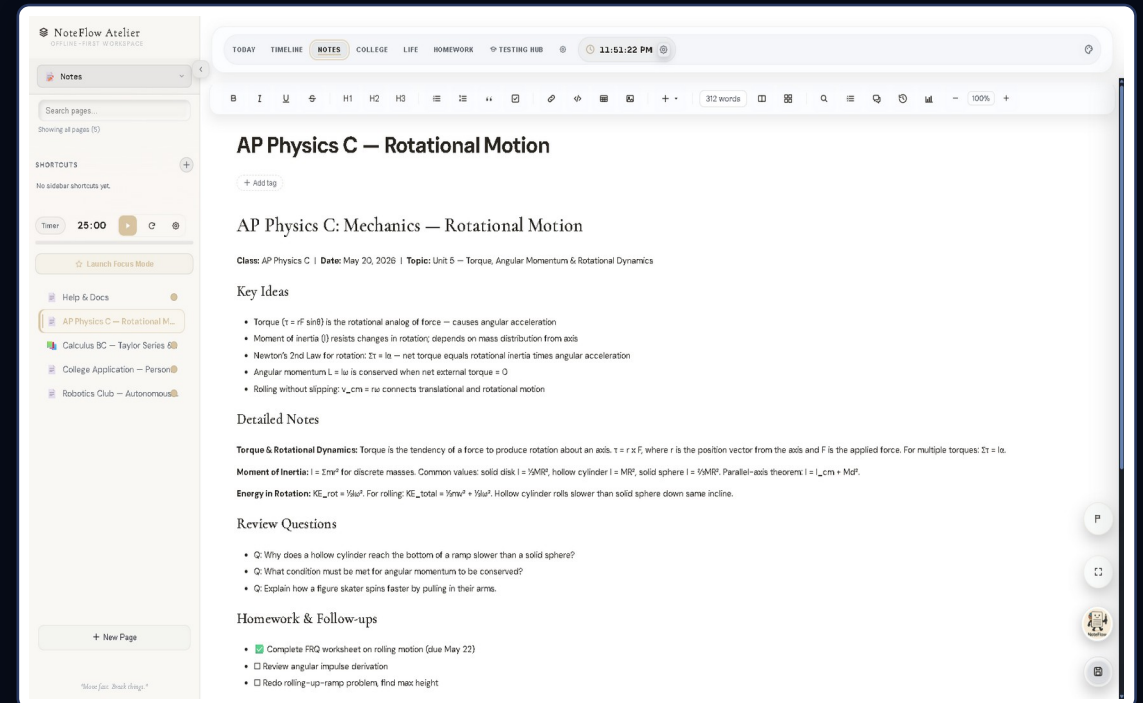
A single top tab switcher exposes the workspace; the active Sutra Mode decides which views are primary.

- Today — daily command center + one Next Step
- Timeline — month/week/day/year planner
- Notes — hierarchical tree + rich editor
- Homework, AP Study, Testing Hub, Review
- College, Life, Projects & Work
- Sutra Assistant launcher, bottom-right
- Settings — themes, customization, data, modes

A writing surface that organizes itself

Available Today

- Hierarchical page tree with :: nesting, search, tags, favorites, drag-reorder
- Rich editor: formatting, slash menu, insert menu, tables, embeds, checklists
- Page Mode, per-page Document Backgrounds, split view
- Handwriting blocks (pen / highlighter / eraser), stored as vectors
- Locked pages — PIN as a salted SHA-256 hash, never raw
- Version history, and full export inside every .sutra backup

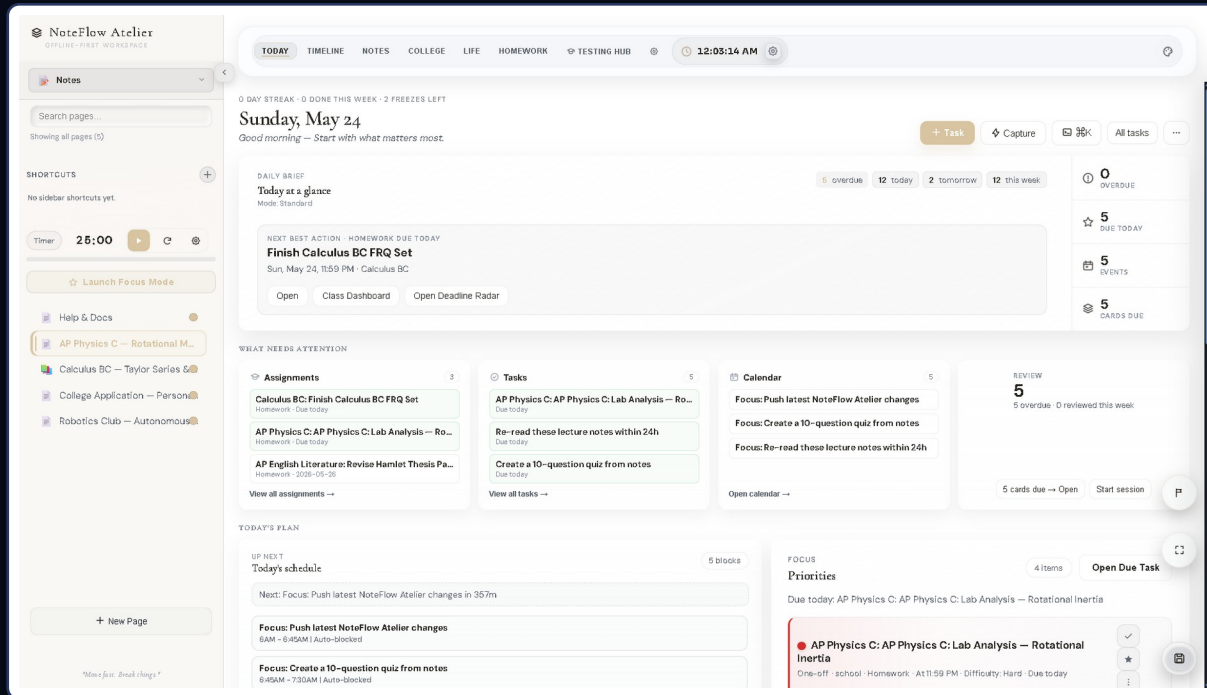


Notes editor — real screenshot (pre-rebrand build)

Every note travels with your workspace and round-trips through backups.

Deadlines, classes, and time in one place

Available Today



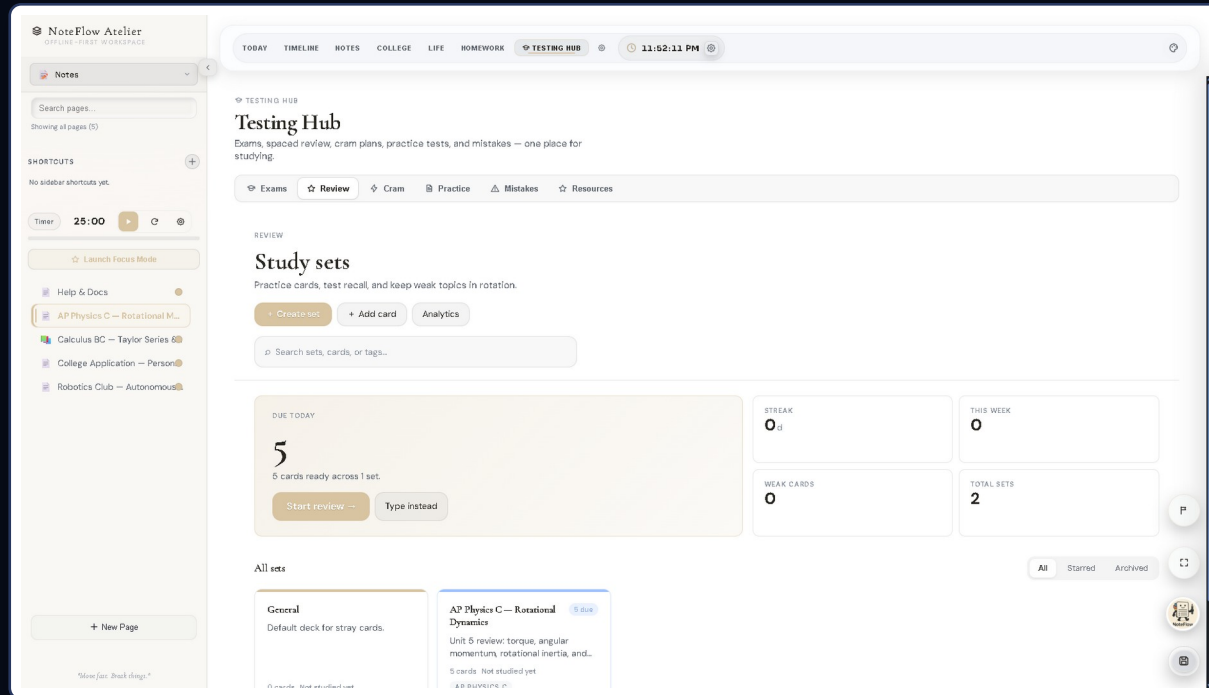
Today command center (pre-rebrand build)

- Today: a Daily Thread + a deterministic single Next Step
- Deadline Radar groups every deadline by urgency, with Schedule this
- Homework lanes for Subjects and Activities, with due-state chips
- Import from School Portal parses pasted portal lines
- Timeline calendar — month/week/day/year, recurrence, ICS in/out
- Quick Capture turns 'Chem essay due Friday hard' into the right item

Studying connected to the work that drives it

Available Today

Testing Hub: In Progress



Testing Hub (pre-rebrand build)

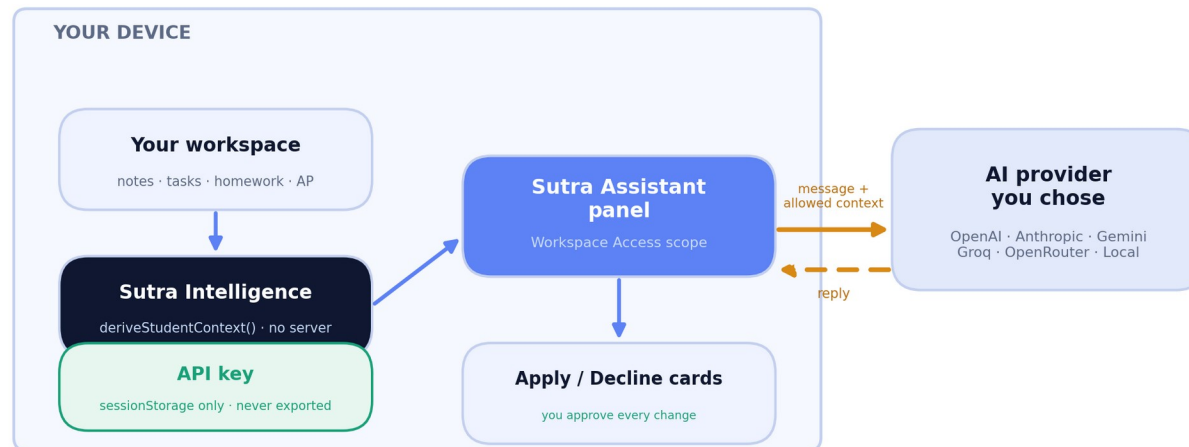
- Review — spaced repetition with a local SM-2-lite scheduler, no AI, no backend
- Five modes: Flashcards, Learn, Write, Test, Match
- AP Study: units, sessions, practice logs, weak-area flags, countdowns
- AP Battle Plan recommends a concrete next session with reasoning
- Testing Hub: dashboard-first pinned exams (being deepened)
- Review due surfaces on Today; decks index into global search

Optional AI that reads your workspace — with your approval

Available Today

Sutra Assistant — Data Flow

Local signals ground every reply. Sutra runs no model servers and no relay.



The "Powered by Sutra Intelligence" badge keeps the privacy boundary visible. Network is reached only when you send a message.

Assistant data flow

- Bring your own key: OpenAI, Anthropic, Gemini, Groq, OpenRouter, or a local endpoint
- Sutra Intelligence derives local signals on-device — calls no server
- Workspace Access: Current Screen / Area / Full
- Single Request vs Conversation Memory
- Apply / Decline cards — you approve every change
- Keys live in `sessionStorage` only; never exported

Local-first by design

1

You own it

Your workspace lives in your browser's storage. No login, no Sutra account.

2

Fast

Every data operation is local — no network round-trip to read or edit.

3

Private

No telemetry, no cloud sync. Nothing is uploaded by the app itself.

4

Resilient

Once loaded, the workspace needs no connection to read or edit.

5

Portable

A single .sutra file is a complete, movable backup of everything.

6

Trustworthy

The privacy boundary is visible and the export is yours to keep.

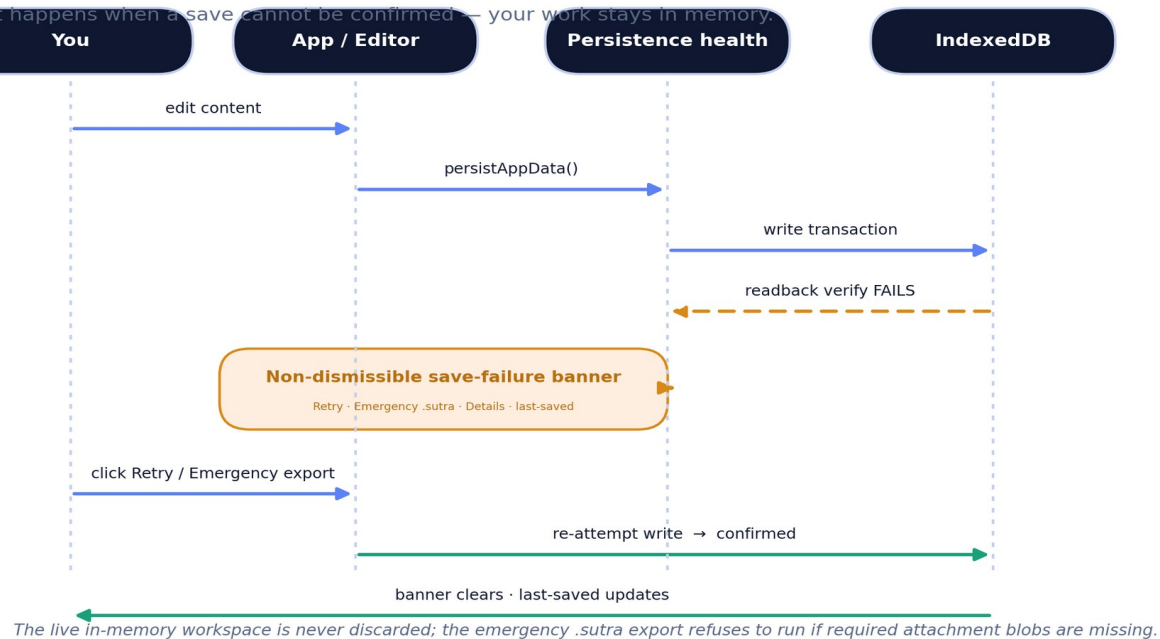
Grounded in the implementation: IndexedDB workspace, debounced save + lifecycle flush, and a vendored ZIP library so core backups make zero third-party requests.

Your work is hard to lose

Available Today

Autosave & Recovery Sequence

What happens when a save cannot be confirmed — your work stays in memory.



Autosave & recovery sequence

- Debounced autosave + a synchronous flush on page-hide / unload
- A persistence-health pipeline classifies quota / serialization / transaction / attachment failures
- A non-dismissible banner with Retry + Emergency .sutra export
- The live in-memory workspace is always preserved
- Pre-import safety snapshot before every import
- Emergency export refuses to run if attachment blobs are missing

Privacy and security posture

● Stays on device

- Entire workspace in IndexedDB + localStorage
- Document backgrounds + inline images local
- No telemetry, no required account

● Leaves only on your action

- AI calls go browser → your chosen provider
- Approved media embeds + AP resource links
- Optional import/export helper libraries

● Never leaves

- API keys (sessionStorage only, redacted)
- Conversation history (session-local)
- Secrets stripped from every export

Known limits: meta-CSP cannot set frame-ancestors (header-only — unavailable on GitHub Pages); script-src uses 'unsafe-inline' for the no-build app; .sutra exports are not encrypted (treat as personal files).

Make it yours — without a marketplace

Available

Themes

21+ presets incl. the Sutra signature, Retro, macOS 26, Windows 11, and more. Apply to current page, all pages, or a custom subset.

Available

CSS Overrides

Named snippets with enable/disable, live preview, brace validation, import/export, and a non-destructive reset. Applied after themes.

In Progress

Plugins

Local bundles only, sandboxed in an iframe behind a permission allowlist. Install disabled; re-review forced on import.

Available

Safe Mode

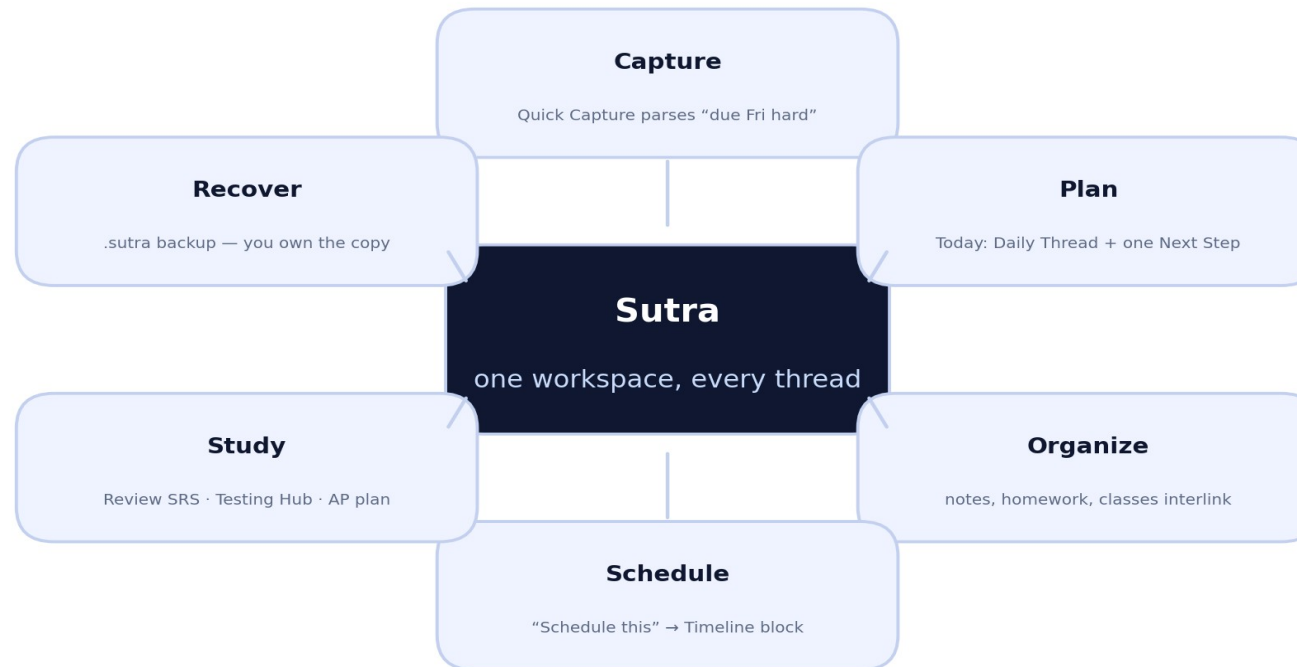
Loads with no custom CSS and no plugins, and never deletes anything — the safe way back from a bad customization.

Everything is local-first and travels inside your workspace backup. No accounts, no store, no server.

A complete student workflow

A Day in Sutra — Continuous Student Workflow

One thread through capture, plan, study, and review — without scattering context.



Where Sutra is different

Capability	Sutra	Notes app	Task mgr	AI chatbot	School LMS	Study app
Student-specific workflow	Yes	~	No	No	Per-school	Partial
Local-first data ownership	Yes	Varies	Varies	No	No	Varies
Academic planning + deadlines	Yes	No	Yes	No	Read-only	No
Notes ↔ study ↔ tasks continuity	Yes	No	No	No	No	No
Spaced-repetition review	Yes	No	No	No	No	Yes
Recoverable single-file backup	Yes	Varies	Varies	No	No	No
Optional, approval-gated AI	Yes	Add-on	No	Core	No	Varies
Deep customization / themes / CSS	Yes	Varies	Limited	No	No	Limited

Categories, not specific competitors. '~' / 'Varies' = depends on the product. Sutra's edge is continuity + ownership, not any single feature.

An honest read on maturity

● Functional today

Notes, Today, Timeline, Homework, AP Study, Review, College, Life, Work, local-first save, .sutra backup, themes — all shipped, with a 14-check release gate passing.

● Being hardened

Mobile QA across every surface, Testing Hub depth, onboarding polish, the plugin ecosystem, and header-level security on the host.

● Needs further QA

The manual viewport × surface matrix and physical-device testing are not yet signed off; the encoding gate suppresses 22 allowlisted findings.

● Experimental / Roadmap

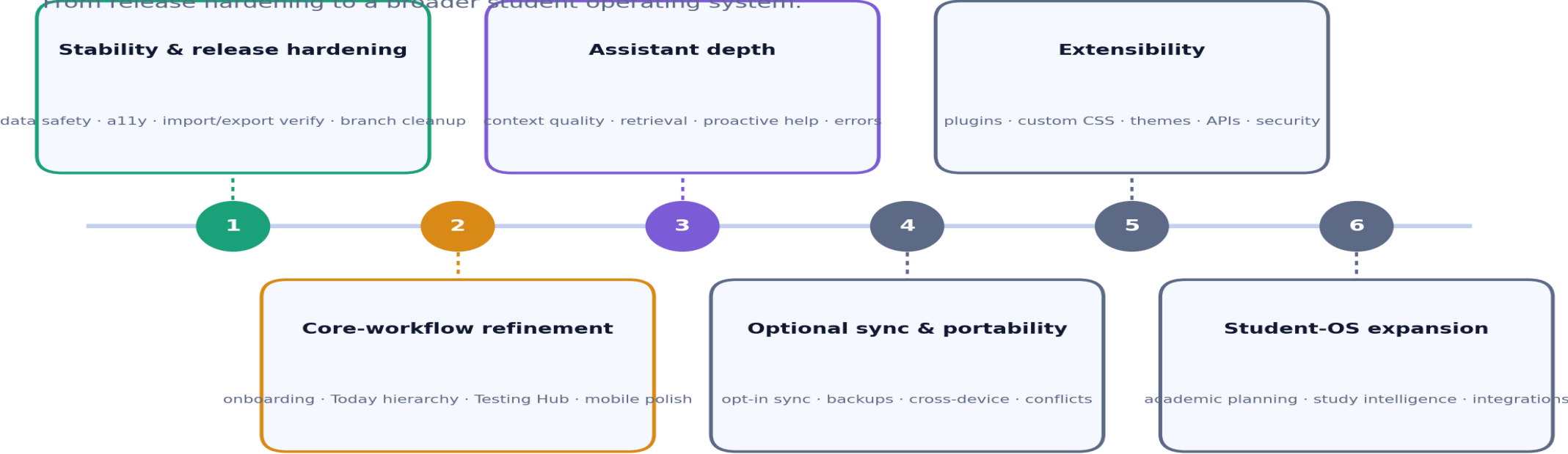
Cram and proactive assistance are light; opt-in sync, a service worker, and cross-device workflow are roadmap, not present.

Verdict: strong enough for private testing, portfolio, and a controlled student beta; not yet production-critical for users with no backups.

Where Sutra goes next

Phased Roadmap

From release hardening to a broader student operating system.



Why Sutra matters

Students don't need another isolated app.

They need a workspace that preserves context, protects their work, and turns academic obligations into organized action — while keeping their data their own.

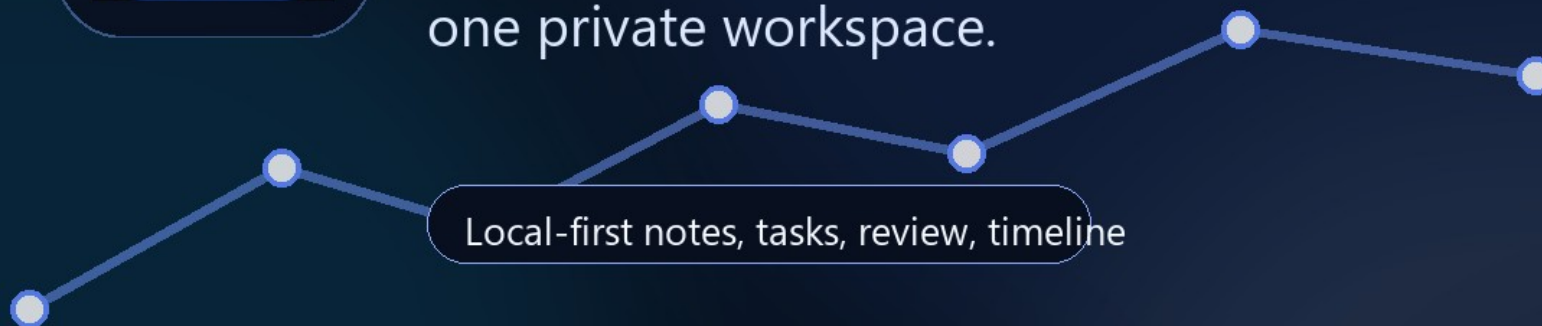


That is the thread Sutra runs through a student's whole academic life.



Sutra

Your academic life, woven into
one private workspace.



Local-first notes, tasks, review, timeline

No account required. Export complete .sutra backups.

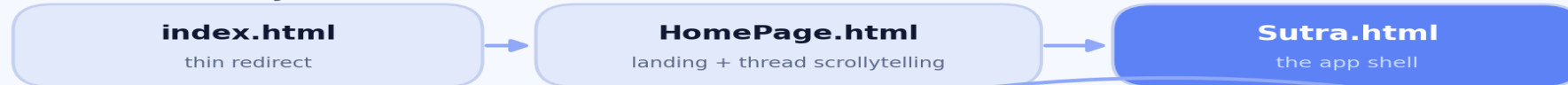
One workspace. Every thread.

Technical architecture

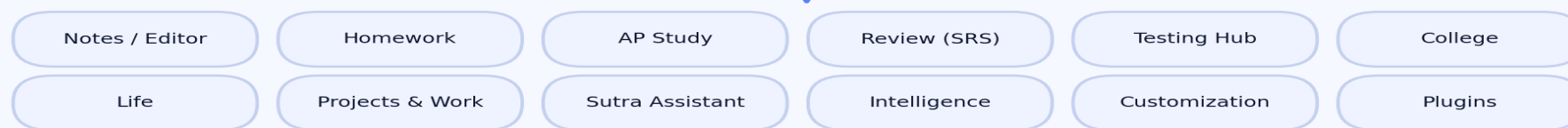
Sutra — High-Level Product Architecture

Static, no-build web app. The page you open is the app. No backend, no bundler.

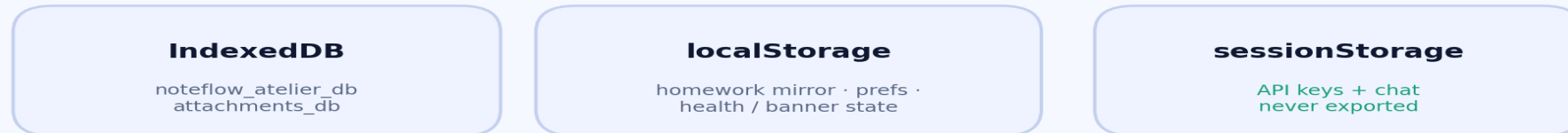
THE BROWSER (your device)



src/features/ · src/ui/ (feature modules attach to the core on window)



LOCAL STORAGE (legacy-named compatibility identifiers retained)

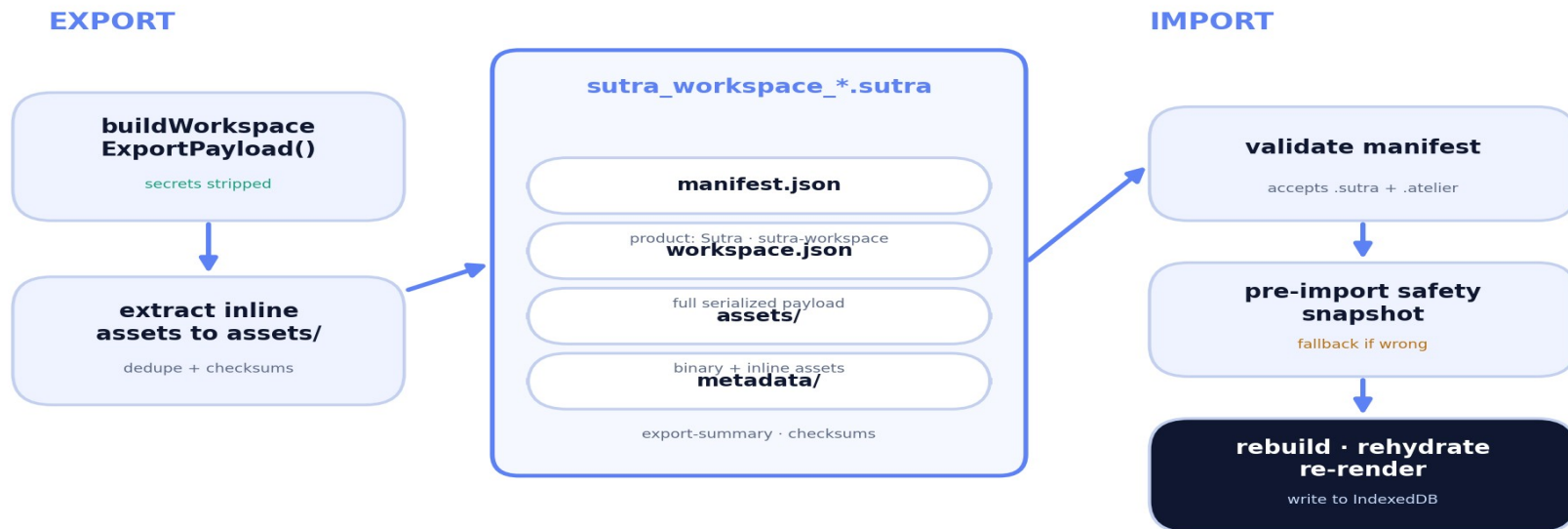


External calls happen ONLY when you trigger them → your chosen AI provider · approved media embeds · optional import/export helpers

.sutra export & import lifecycle

.sutra Export & Import Lifecycle

A .sutra file is a ZIP package — a complete, portable copy of your workspace.

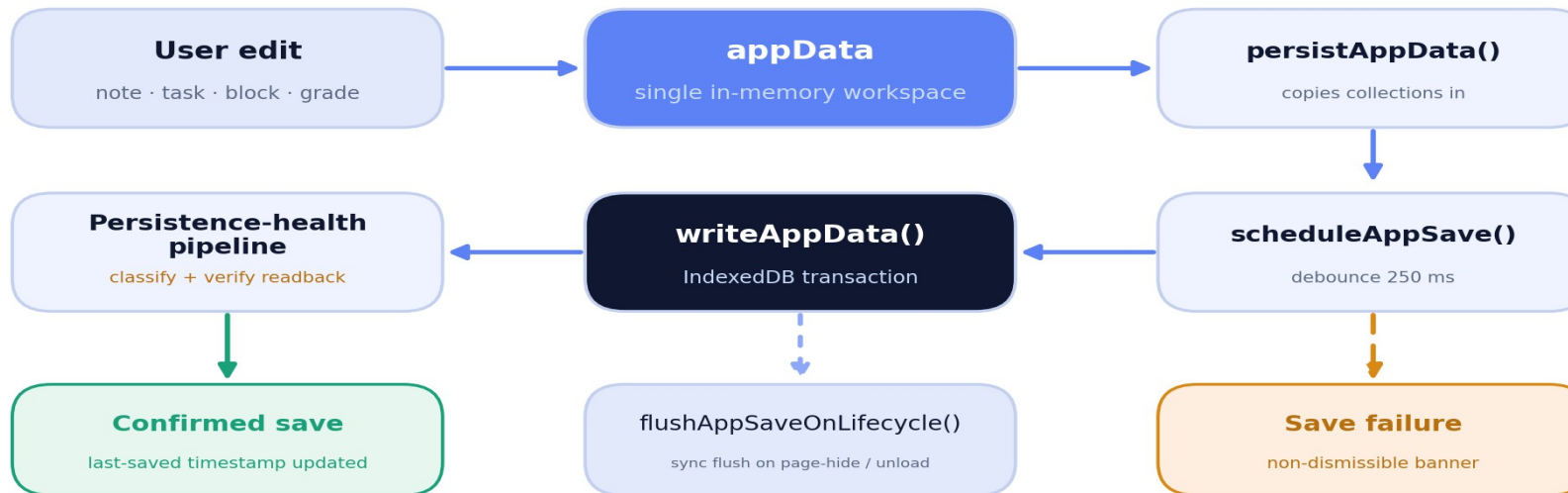


Excluded by design: API keys / secrets (redacted), conversation history, regenerable caches, ephemeral UI state. Provider/model CHOICES travel; the key is re-entered.

Persistence flow

Local-First Persistence Flow

One in-memory workspace object, one hydrate path, one debounced save path.



Failure classes: quota · serialization · IndexedDB transaction · attachment · cache-warming · partial-write/readback. The in-memory workspace is always preserved.

Release-readiness summary

Audience	Ready?	Notes
Private testing	Yes	All 14 static guards pass; persistence + recovery verified.
Portfolio / showcase	Yes	Strong story; recapture Sutra-branded screenshots first.
Hackathon / demo	Yes	Local-first means no server to fail on stage.
Controlled student beta	Yes, with caveats	Walk the device QA matrix; fix the 'Flowy' string.
Wider public beta	Almost	Move to a header-capable host (frame-ancestors); finish mobile QA.
Production-critical use	Not yet	No sync/service worker; users must keep .sutra backups.